



## Reflect with us - Week 7



This assignment will not be graded and is only for practice.

1 point

Let  $L_1$  and  $L_2$  be affine subspaces of  $R^3$ , where  $L_1 = U$  and  $L_2 = (2, 0, 1) + W$ , for some vector subspaces  $U$  and  $W$  of  $R^3$ . Let a basis for  $U$  be given by  $\{(2, 0, 1), (1, 1, 0), (0, 1, 0)\}$  and a basis for  $W$  be given by  $\{(1, 0, 1), (0, 1, 1)\}$ . Suppose there is a linear transformation  $T: U \rightarrow W$  such that  $(0, 1, 0) \in \ker(T)$ ,  $T(2, 0, 1) = (0, 1, 1)$  and  $T(1, 1, 0) = (1, 0, 1)$ . An affine mapping  $f: L_1 \rightarrow L_2$  is obtained by defining  $f(u) = (2, 0, 1) + T(u)$ , for all  $u \in U$ . Which of the following options are true?

☐

$$L_1 = R^3$$

☐

$$L_2 = \{(x, y, z) \mid x + y - z = 1\}$$

☐

$$L_2 = R^3$$

☐

$$f(x, y, z) = (x - 2z + 2, z, x - z + 1)$$

☐

$$f(x, y, z) = (x - 2z, z, x - z)$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$L_1 = R^3$$

$$L_2 = \{(x, y, z) \mid x + y - z = 1\}$$

$$f(x, y, z) = (x - 2z + 2, z, x - z + 1)$$

**Solution:**

Discussion on Option 1: It is given that  $L_1 = U$  and  $\{(2, 0, 1), (1, 1, 0), (0, 1, 0)\}$  is a basis of the vector subspace  $U$ .

The dimension of  $U$  is

Hint

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 3

1 point

$L_1 = U$  is a subspace of  $R^3$ . To know whether  $L_1 = U$  is a proper subspace



$R^3$  or not, we just have to compare their dimensions.

1 point

Now, can you see why  $L_1 = U = R^3$   $L_1 = U = R^3$  ?

☐ Yes

☐ No

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

**Accepted Answers:**

Yes

Discussion on Options 2 and 3: Discussion on Options 2 and 3: It is given that  $L_2 = (2, 0, 1) + W$   $L_2 = (2, 0, 1) + W$  and  $\{(1, 0, 1), (0, 1, 1)\}$   $\{(1, 0, 1), (0, 1, 1)\}$  is a basis for the vector subspace  $W$ . Now to find the explicit description of  $L_2$ , we have to find the explicit description of  $W$  first.

$$W = \{a(1, 0, 1) + b(0, 1, 1) \mid a, b \in R\} \\ = \{(a, b, a + b) \mid a, b \in R\}$$

$$\{a(1, 0, 1) + b(0, 1, 1) \mid a, b \in R\} = \{(a, b, a + b) \mid a, b \in R\}$$

As  $W$  is a subspace of  $R^3$ , we want to express it using the usual  $x, y, z$  co-ordinate and the relation between them. From the above description we have,  $x = a$ ,  $y = b$ ,  $z = a + b$ , which implies that  $z = x + y$ . So

$$W = \{(x, y, z) \mid x + y - z = 0, \text{ where } x, y, z \in R\} \\ = \{(x, y, z) \mid x + y - z = 0, \text{ where } x, y, z \in R\}$$

1 point

Now, are you able to find the equation of  $L_2$ ?

☐ Yes

☐ No

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

**Accepted Answers:**

Yes

1 point

Do you agree that the sets  $\{(2 + x, y, z + 1) \mid x + y - z = 0\}$  and  $\{(a, b, c) \mid a + b - c = 1\}$  are the same?

☐ Yes

☐ No

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

**Accepted Answers:**

Yes

## Discussion on Option 4 and 5: Discussion on Option 4 and 5:

Step 1: Step 1: Here, we have to find an algebraic expression for the affine map  $f: L_1 \rightarrow L_2$ . To get an algebraic expression for  $f$ , we have to find the algebraic expression which represents the linear transformation  $T: U \rightarrow W$ .

Given:

$$T(0, 1, 0) = (0, 0, 0), \quad T(2, 0, 1) = (0, 1, 1), \quad T(1, 1, 0) = (1, 0, 1)$$

1 point

Is  $T$  uniquely determined by the above information?

☐ Yes

☐ No

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

### Accepted Answers:

Yes

Step 2: Step 2: Since  $\{(0, 1, 0), (2, 0, 1), (1, 1, 0)\}$  is a basis of  $U = \mathbb{R}^3$ , we can write any element  $(x, y, z) \in U$  as a linear combination of  $\{(0, 1, 0), (2, 0, 1), (1, 1, 0)\}$ .

$$(1) \quad (x, y, z) = a(0, 1, 0) + b(2, 0, 1) + c(1, 1, 0)$$

Apply  $T$  on both sides:

$$T(x, y, z) = aT(0, 1, 0) + bT(2, 0, 1) + cT(1, 1, 0)$$

$$T(x, y, z) = a(0, 0, 0) + b(0, 1, 1) + c(1, 0, 1)$$

To get an algebraic expression for  $T(x, y, z)$ , we just have to know the values of  $a$ ,  $b$  and  $c$  in terms of  $x$ ,  $y$  and  $z$ .

1 point

Now, can you find the values of  $a$ ,  $b$  and  $c$  using equation (1)?

☐ Yes

☐ No

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

### Accepted Answers:

Yes

Solve the above system for  $a$ ,  $b$  and  $c$ .

1 point

What is the value of  $a$ ?

☐

$z$

☐

$x - 2zx - 2z$

☐

$y + 2z - xy + 2z - x$

☐

$x - yx - y$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$y + 2z - xy + 2z - x$

*1 point*

What is the value of  $bb$  ?

☐

$zz$

☐

$x - 2zx - 2z$

☐

$y + 2z - xy + 2z - x$

☐

$x - yx - y$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$zz$

*1 point*

What is the value of  $cc$  ?

☐

$zz$

☐

$x - 2zx - 2z$

☐

$y + 2z - xy + 2z - x$

☐

$x - yx - y$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$x - 2zx - 2z$

Step 3:Step 3:

$$T(x, y, z) = (0, z, z) + (x - 2z, 0, x - 2z) = (x - 2z, z, x - z) \quad T(x, y, z) = (0, z, z) + (x - 2z, 0, x - 2z) = (x - 2z, z, x - z)$$

By definition:

$$f(x, y, z) = (2, 0, 1) + T(x, y, z) \quad \text{where } (x, y, z) \in U. \quad f(x, y, z) = (2, 0, 1) + T(x, y, z) \quad \text{where } (x, y, z) \in U.$$

By substituting the value  $T(x, y, z)$  in the above expression, we get

$f(x, y, z) = (x - 2z + 2, z, x - z + 1)$  for all  $(x, y, z) \in U$ .  $f(x, y, z) =$   
 $(x - 2z + 2, z, x - z + 1)$  for all  $(x, y, z) \in U$ .

[Check Answers](#)